**INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA**
**DEPARTMENT OF MECHATRONICS ENGINEERING**

MECHATRONIC SYSTEM INTEGRATION (MCTA 3203)
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# EXPERIMENT 1: DIGITAL LOGIC DESIGN

SECTION: 1 | GROUP: 11
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| NO. | MATRIC NO. | NAME |
|---|---|---|
| 1 | 2414273 | DANISH IMAN BIN AJO |
| 2 | 2414585 | ALIFF EMIR BIN AHMAD EZAD |
| 3 | 2416879 | AHMAD AMIRULAQIL BIN ZAINORKHAIR |
| 4 | 2418895 | MUHAMMAD IRFAN BIN ROSHDAN |

## ABSTRACT

The objective of this experiment is to build a manual sequential counter with reset primarily using Arduino UNO as the microcontroller alongside various components such as push buttons, resistors and a 7-segment display. The procedure of this experiment starts off with connecting the 7-segment display to the Arduino UNO by first connecting it to the breadboard and hooking up each one of its legs to an unoccupied digital pin with a resistor in between to limit the current flow. Then, two push buttons are connected to the Arduino UNO with two of its legs going to 5V and GND and one more leg connected to an unoccupied digital pin for each button to act as a counter button and a sequence reset button. The Arduino UNO can be coded accordingly to achieve a proper result while also being wary of the errors we have come across which is an issue with the 7-segment display not displaying the desired numbers and it being a LOW input device. These issues were relatively easy to resolve and it proved to be not a huge obstacle in accomplishing our project of manual sequential counter with reset. Various recommendations can be made in order to flesh out this project further, such as a second 7-segment display could be added to make a two digit counter and adding a buzzer and an LED indicator to enhance user feedback.

## TABLE OF CONTENTS

**ABSTRACT** ..... 2
**TABLE OF CONTENTS** ..... 2
**PART 1** ..... 2
- **1.0 INTRODUCTION & THEORY** ..... 2
- **1.1 MATERIALS AND EQUIPMENT** ..... 2
- **1.2 EXPERIMENTAL SETUP** ..... 3
- **1.3 METHODOLOGY & PROGRAMMING LOGIC** ..... 4
- **1.4 DATA COLLECTION & ANALYSIS** ..... 8
- **1.5 RESULTS & DISCUSSION** ..... 9
- **1.6 CONCLUSION** ..... 9

**RECOMMENDATIONS** ..... 10
**REFERENCES** ..... 10
**APPENDICES** ..... 10
**ACKNOWLEDGEMENT** ..... 10
**STUDENT DECLARATION** ..... 11

## PART 1

### 1.0 INTRODUCTION & THEORY

Arduino UNO is a microcontroller used for a variety of applications in the mechatronics world. The Arduino UNO is easily programmable and usable with many electrical and electronic components which allows for projects and systems to be built quickly. In this experiment, components that were

used are push buttons, a 7-segment display, and 220-ohm resistors which accomplishes a manual sequential counter with reset. A push button is wired with one of its legs connected to an unoccupied digital pin (e.g. D7), another one connected to 5V and a third leg connected to GND with a resistor in between. The D7 digital pin will now read the states of the push button which will in turn be used in coding as an input for another component. Two of these push button assemblies will be used as the counter and reset buttons. The 7-segment display is an LED display with 7 individual segments that can be turned on and off separately. Each one of the segments' legs has to be connected to an unoccupied digital pin with a resistor in between to limit current flow. By using these components together, we can accomplish a manual sequential counter with reset.

## 1.1 MATERIALS AND EQUIPMENT

- Microcontroller (e.g., Arduino Mega 2560)
- Breadboard & Jumper Wires
- PC with arduino IDE Environment\
- Resistor 10k , 220k
- 7 segment display
- buttons

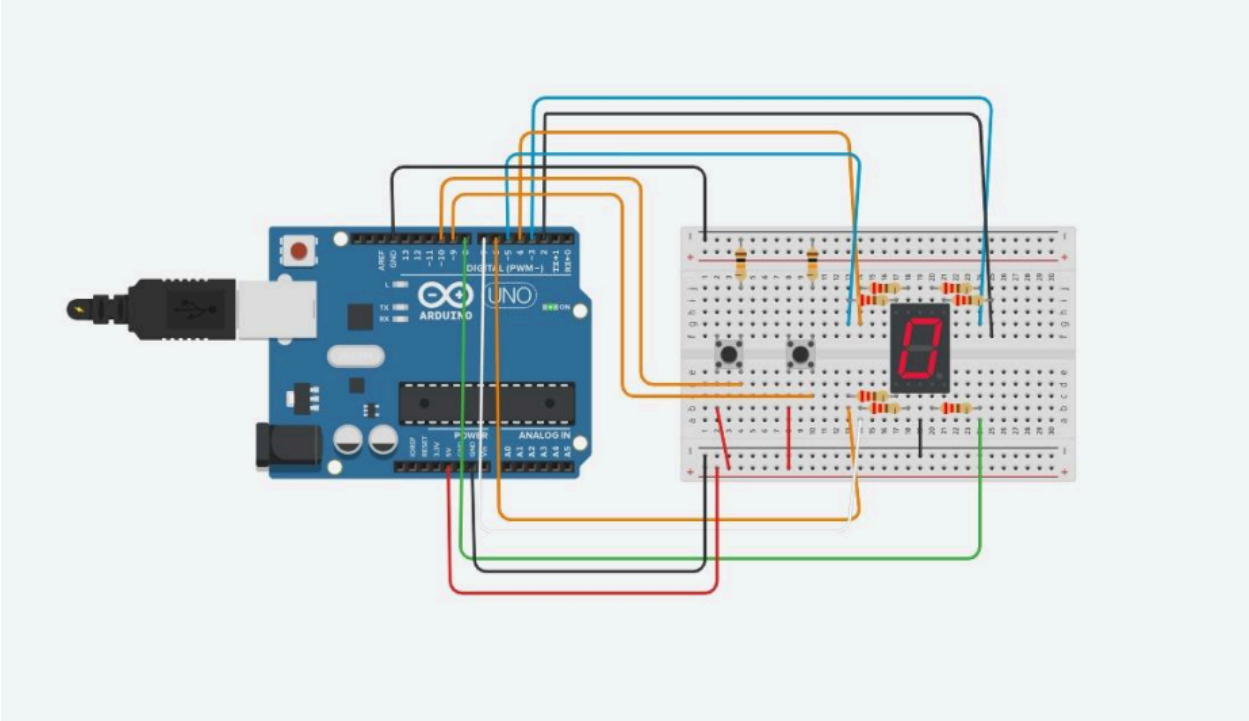
## 1.2 EXPERIMENTAL SETUP

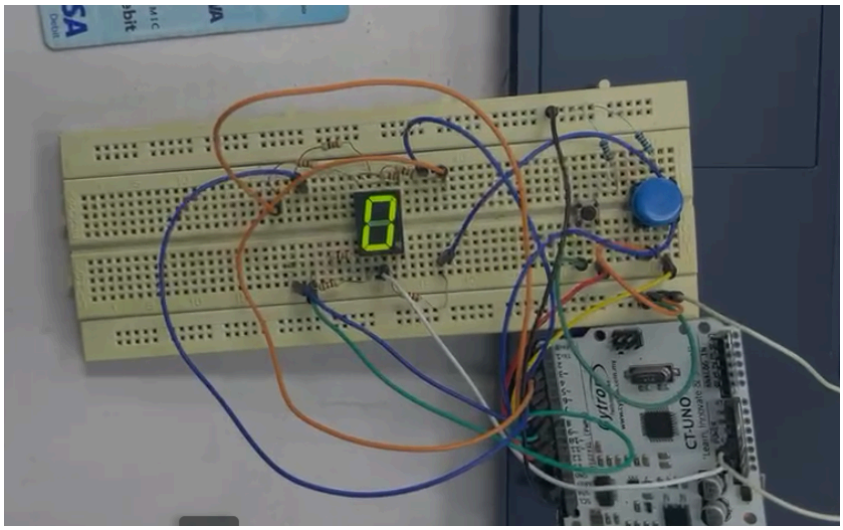
1. Connect the common cathode 7-segment display to the Arduino Uno as follows:

- Connect each of the 7 segments (a, b, c, d, e, f, g) of the display to separate digital pins on the Arduino (e.g., D0 to D6).
- Connect the common cathode pin of the display to one of the GND (ground) pins on the Arduino.
- Use 220-ohm resistors to connect each of the segment pins to the Arduino pins to limit the current.

2. Connect the push buttons to the Arduino:

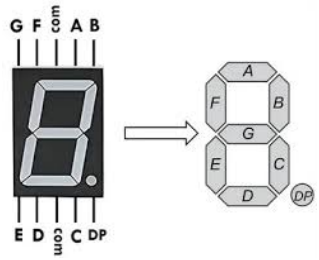
- Connect one leg of each push button to a separate digital pin (e.g., D9 and D10) and connect the other leg of each push button to GND.
- Use 10K-ohm pull-up resistors for each push button by connecting one end of each resistor to the digital pin and the other end to the 5V output of the Arduino.



## 1.3 METHODOLOGY & PROGRAMMING LOGIC

Each segment is led and needs an instruction for it to light up. This display contains seven segments and needs to instruct each segment independently to light it up. For this particular display the segment led will turn on when we send a low signal which is 0 into it. We set specific segments to low to produce numbers from 0 to 9 and by using the button we can increase the count by one.




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For example to make a zero A = 0 ,  B = 0 , C = 0, D = 0, E = 0, F = 0.

Here is the full code:

```
const int segmentA = 3;

const int segmentB = 2;

const int segmentC = 8;

const int segmentD = 7;

const int segmentE = 6;

const int segmentF = 4;

const int segmentG = 5;

const int button_1 = 9;

const int button_2 = 10;


int count = 0;

int num_array[10][7] = {

{0, 0, 0, 0, 0, 0, 1},//0

{1, 0, 0, 1, 1, 1, 1},//1

{0, 0, 1, 0, 0, 1, 0},//2

{0, 0, 0, 0, 1, 1, 0},//3

{1, 0, 0, 1, 1, 0, 0},//4

{0, 1, 0, 0, 1, 0, 0},//5

{0, 1, 0, 0, 0, 0, 0},//6
```

```
{0, 0, 0, 1, 1, 1, 1},//7
{0, 0, 0, 0, 0, 0, 0},//8
{0, 0, 0, 0, 1, 0, 0}//9

};
void setup() {
pinMode(segmentA, OUTPUT);
pinMode(segmentB, OUTPUT);
pinMode(segmentC, OUTPUT);
pinMode(segmentD, OUTPUT);
pinMode(segmentE, OUTPUT);
pinMode(segmentF, OUTPUT);
pinMode(segmentG, OUTPUT);
pinMode(button_1, INPUT);
pinMode(button_2, INPUT);
display(num_array[count]);

}
void loop() {
if (digitalRead(button_1) == HIGH) {
if (count < 9) {
count++;
}
else {
count = 0;//number 0
}
```

```
display(num_array[count]);
while (digitalRead(button_1) == HIGH) {
delay(10);
}
delay(50);
}

else if (digitalRead(button_2) == HIGH) {
count = 0;
display(num_array[count]);
while (digitalRead(button_2) == HIGH) {
delay(10);
}
delay(50);
}
}
void display(int bin[]) {
digitalWrite(segmentA, bin[0]);
digitalWrite(segmentB, bin[1]);
digitalWrite(segmentC, bin[2]);
digitalWrite(segmentD, bin[3]);
digitalWrite(segmentE, bin[4]);
digitalWrite(segmentF, bin[5]);
digitalWrite(segmentG, bin[6]);
}
```

## 1.4 DATA COLLECTION & ANALYSIS

| Bil | Input | Observed Output |
|---|---|---|
| 1 | The arduino is turned on | The 7-segment displays '0' |
| 2 | The pushbutton_1 is pressed | The 7-segment displays value increment to '1' |
| 3 | The pushbutton_1 is pressed | The 7-segment displays value increment to '2' |
| 4 | The pushbutton_1 is pressed | The 7-segment displays value increment to '3' |
| 5 | The pushbutton_1 is pressed | The 7-segment displays value increment to '4' |
| 6 | The pushbutton_1 is pressed | The 7-segment displays value increment to '5' |
| 7 | The pushbutton_1 is pressed | The 7-segment displays value increment to '6' |

| | | |
|---|---|---|
| 8 | The pushbutton_1 is pressed | The 7-segment displays value increment to '7' |
| 9 | The pushbutton_1 is pressed | The 7-segment displays value increment to '8' |
| 10 | The pushbutton_1 is pressed | The 7-segment displays value increment to '9' |
| 11 | The pushbutton_1 is pressed | The 7-segment displays value increment to '0' |
| 12 | The pushbutton_2 is pressed | The 7-segment displays '0' |

## 1.5 RESULTS & DISCUSSION

When using the seven segment display using the code given in the module, we notice a slight error where the segment doesn't show the numbers properly. We manage to fix the problem by changing which segment should be lit up for specific numbers that need to be shown. Another problem that we encounter is that this specific seven segment display requires a low input which is zero for the led to light up. After several iterations we managed to make the seven segments work and increase the increment by one when the button is pressed and reset to zero when the other button is pressed.

## 1.6 CONCLUSION

This project successfully demonstrated the design and implementation of a seven-segment display system using an Arduino microcontroller. The system was able to display numerical digits from 0 to 9 accurately by controlling the output signals sent to each segment. After going through the process of putting the circuit together, coding it and testing, the project achieved its main objective that is - understand and implement digital logic interfacing between Arduino and external display devices, 7-segment display. In terms of understanding, we now know how to code a counting machine and as for implementing, we are confident in the reenactability of this project. The results proved that even simple electronic systems can be effectively developed through systematic design and logical

programming. For an Islamic reflection, this project highlights the importance of seeking knowledge (Ilm') seen in our objective to understand its inner workings. Knowledge is a gift from Allah that enables mankind to understand and control the world around them. By exploring the principles of electronics and control systems, we recognize the precision and order that Allah has placed in creation.

## RECOMMENDATIONS

1. Improve Display Range

Extend the display range to include two 7-segment displays to show two-digit numbers (00-99). This can be achieved by using a multiplexer or by connection and additional display with separate control pin.

2. Enhance User Interface

Add more push buttons for more additional functions such as decrement, pause, or auto-count mode to make the system more interactive.

3. Add Light or Sound Indicators

Connect a buzzer or an LED to indicate a button is pressed or when the counter exceeds certain limits or when it resets to zero for better user feedback.

4. Decimal Value

With a potentiometer and two 7-segment displays connected to show decimal points value to experiment with coding to change analog value received from the potentiometer to a digital output.

## REFERENCES

Prabhu, A. (2026, January 20). *How seven segment display works & interface it with Arduino*. Last Minute Engineers. https://lastminuteengineers.com/seven-segment-arduino-tutorial/

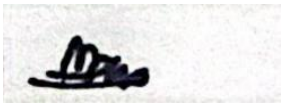
## ACKNOWLEDGEMENT

Alhamdulillah, all praise and gratitude be to Allah SWT for His blessings, guidance, and mercy that enabled us to successfully complete this project entitled "Digital Logic System." Throughout the development process, we faced several challenges, but with His will and guidance, we managed to overcome them and complete the project successfully. We would also like to express our heartfelt appreciation to our lecturer for their continuous guidance, support, and valuable feedback throughout this project. Their advice has greatly helped us in understanding both the theoretical and practical aspects of microcontroller systems and digital display design.

## STUDENT DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgments, and that the original work contained herein has not been undertaken or done by unauthorized sources or persons.

| SIGNATURE | STUDENT DETAILS | CHECKLIST |
|---|---|---|
|  | Name: DANISH IMAN BIN AJO<br>Matric No: 2414273 | Read [ ✓ ]<br>Understand [ ✓ ]<br>Agree [ ✓ ] |
|  | Name: ALIFF EMIR BIN AHMAD EZAD<br>Matric No: 2414585 | Read [ / ]<br>Understand [ / ]<br>Agree [ / ] |
|  | Name:AHMAD AMIRULAQIL BIN ZAINORKHAIR<br>Matric No: 2416879 | Read [ ✓ ]<br>Understand [ ✓ ]<br>Agree [ ✓ ] |
|  | Name: MUHAMMAD IRFAN BIN ROSHDAN<br>Matric No: 2418895 | Read [ / ]<br>Understand [ / ]<br>Agree [ / ] |